Algorithm 1 AutoDesign meta-harness optimization

Require: Fixed model π_θ; initial design harness H_0; evaluator R_{meta}
Require: Training task set $\mathcal{D}_{\text{train}}$; development task set $\mathcal{D}_{\text{dev}}$; outer-loop iterations T
Ensure: Optimized design harness H_T and optimization record $\mathcal{L}$

Run H_0 on $\mathcal{D}_{\text{train}}$; collect execution trajectories $\boldsymbol{\tau}_0$ and evaluator scores $\boldsymbol{s}_0$
Run H_0 on $\mathcal{D}_{\text{dev}}$ and collect evaluator scores $\boldsymbol{s}_0^{\text{dev}}$
for $t = 0$ to $T - 1$ **do**
  The meta-harness optimizer P inspects $\boldsymbol{\tau}_t$, $\boldsymbol{s}_t$, and $\mathcal{L}$ and proposes a candidate updated harness H'_{t+1}
  Run H'_{t+1} on $\mathcal{D}_{\text{train}}$; collect execution trajectories $\boldsymbol{\tau}'_{t+1}$ and evaluator scores $\boldsymbol{s}'_{t+1}$
  Run H'_{t+1} on $\mathcal{D}_{\text{dev}}$ and collect evaluator scores $\boldsymbol{s}'^{\text{dev}}_{t+1}$
  if $J_{\text{train}}(H'_{t+1}) > J_{\text{train}}(H_t)$ **and** $J_{\text{dev}}(H'_{t+1}) \geq J_{\text{dev}}(H_t)$ **then**
    $d_t \leftarrow$ ACCEPT
    $(H_{t+1}, \boldsymbol{\tau}_{t+1}, \boldsymbol{s}_{t+1}, \boldsymbol{s}^{\text{dev}}_{t+1}) \leftarrow (H'_{t+1}, \boldsymbol{\tau}'_{t+1}, \boldsymbol{s}'_{t+1}, \boldsymbol{s}'^{\text{dev}}_{t+1})$
  else
    $d_t \leftarrow$ REJECT
    $(H_{t+1}, \boldsymbol{\tau}_{t+1}, \boldsymbol{s}_{t+1}, \boldsymbol{s}^{\text{dev}}_{t+1}) \leftarrow (H_t, \boldsymbol{\tau}_t, \boldsymbol{s}_t, \boldsymbol{s}^{\text{dev}}_t)$
  end if
  Append the harness checkpoint and iteration record to $\mathcal{L}$
end for
return $(H_T, \mathcal{L})$

accepted only when its performance on $\mathcal{D}_{\text{train}}$ improves and its performance on $\mathcal{D}_{\text{dev}}$ does not decline:

$$\text{Accept}(H'_{t+1}) \iff J_{\text{train}}(H'_{t+1}) > J_{\text{train}}(H_t) \ \wedge \ J_{\text{dev}}(H'_{t+1}) \geq J_{\text{dev}}(H_t). \tag{6}$$

If the condition holds, the meta-harness promotes H'_{t+1} to H_{t+1}; otherwise H_t is retained. Results on the development set are used exclusively by the acceptance gate and are never exposed to P when constructing an update proposal. Therefore, $\mathcal{D}_{\text{dev}}$ serves as a guard against overfitting the harness to the training tasks.

After the acceptance decision, the meta-harness appends the completed iteration to the optimization record $\mathcal{L}$. For each iteration t, $\mathcal{L}$ stores the harness H_t, the trajectories and scores, the selected harness component, the update plan and the corresponding code changes, and the acceptance decision, with a repository checkpoint preserving the harness implementation at that iteration. Trajectories and scores from the development set are not included in the record.

The updated $\mathcal{L}$ is supplied to P as persistent context in the next iteration. When a candidate is rejected, the record allows the next iteration to propose a different update while retaining the evidence of what has already been tried. $\mathcal{L}$ thus supports comparison, reproducibility, and rollback across iterations. Note that the outer loop maintains a single active harness at each iteration and does not perform tree search over harness variants. The complete meta-harness optimization procedure is summarized in Algorithm 1.

Human-in-the-Loop. AutoDesign also supports an optional human-in-the-loop mode, as illustrated in Figure 3. Human intervention can operate through two channels. First, at iteration t, a user may provide directional guidance g_t in natural language, which is supplied to the planner alongside the trajectories and evaluation scores, so that the update proposal becomes $H'_{t+1} = P(H_t, \boldsymbol{\tau}_t, \boldsymbol{s}_t, \mathcal{L}, g_t)$. We introduce this mechanism because the coding agent acting as P may converge prematurely to a locally satisfactory harness configuration, at which point outer-loop optimization stagnates. Guidance can inject task-specific heuristics or redirect the search toward alternative improvements.

Second, human guidance may be provided to the coding agent responsible for implementing the evaluator when visual inspection reveals a systematic artifact bias not captured by R_{meta}. Such evaluator revision requires explicit human input. Otherwise, R_{meta} remains fixed because the meta-harness receives no external signal with which to identify or correct evaluator bias.

In both cases, the human provides observations or high-level directions rather than directly editing the harness or evaluator implementation. When no guidance is provided, the outer loop operates autonomously as defined in Equation (5).